The skipped-curve commonality search: small conductors and completed twists

Date: 2026-10-23 UTC. **Status:** Bounded, retrospective structural search. **Question:** If the original 1997–98 program evaluated all eleven reported qualifying but unpublished seed blocks, do they share a simple arithmetic trait absent from the ten published prime-field curves?

Result and inference boundary

Derived: The fixed 21-feature family in PROTOCOL.md has no exact separator of all published and passed-over labels, either by one global numeric threshold or by strict ordering within every field. In particular, small squared prime factors of the Frobenius discriminant do **not** mark the eleven omissions: just one omitted curve has such a factor at or below 10,000, while two published controls do. This closes a specific small-prime conductor proxy, not an arbitrary endomorphism, isogeny, or hidden-route hypothesis. The strongest association in this bounded family remains the previously noticed low coefficient-`b` bit weight among published curves. It has counterexamples and no established cryptanalytic interpretation.

Cross-checked extension: The two quadratic-twist factorizations left censored by the earlier 20-second probe—published prime239v2 and omitted 239-bit block 2682—completed under the separately frozen 300-second limits. PARI/GP `isprime` returned 1 for every new factor; Sage's proof-enabled primality check and separately recomputed full-order multiplication passed. The two primality calls may share underlying libraries, so this is a second arithmetic check rather than independent primality-algorithm evidence. Together with the earlier 18 completed probes and the later P-224 block-1485 follow-up, the named 21-curve set now has complete recorded twist-order factorizations. The new values do not restore a rule that prefers the largest twist prime factor: older cross-field counterexamples remain.

The user's working premise—that the historical program actually evaluated all eleven and rejected them for a weakness—is stipulated here, **not observed**. The surviving seed streams do not show the original call trace, point-count state, or rejection reason. Neither an arithmetic label split nor a twist factorization demonstrates a weakness in the stated prime-order subgroup.

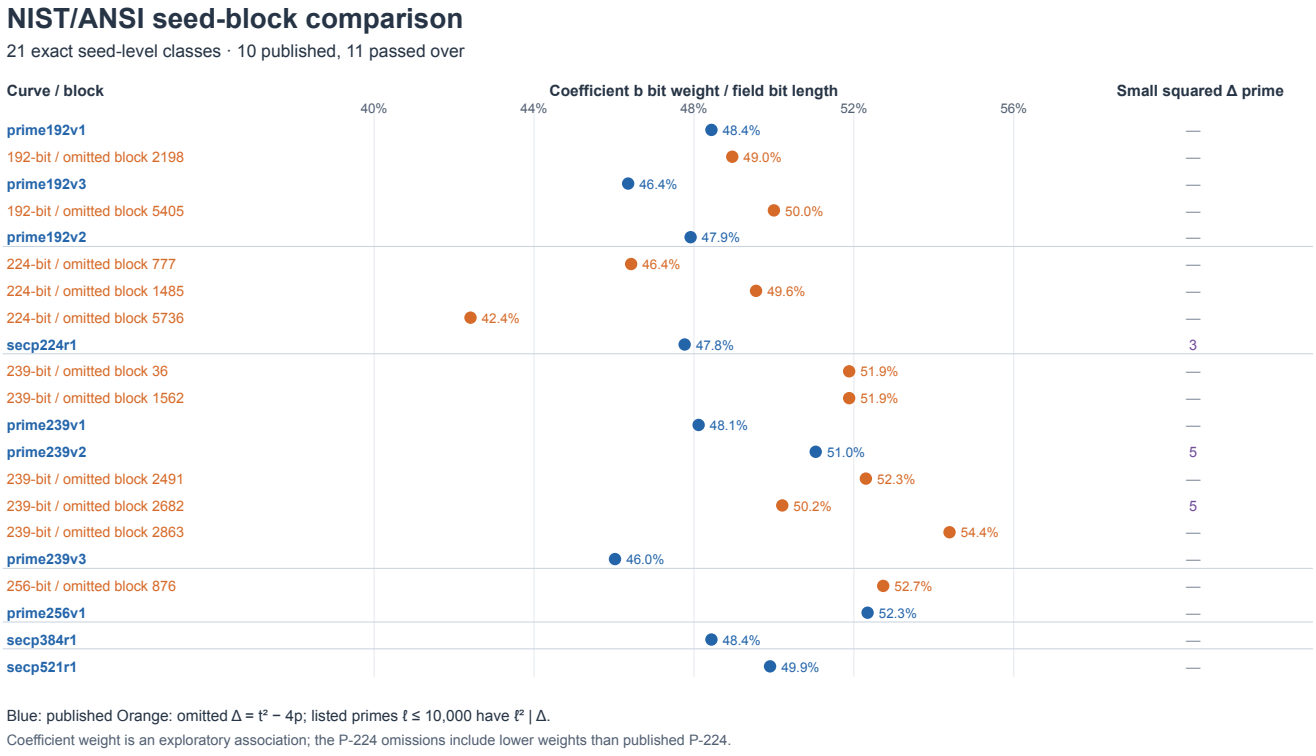


Figure 1: Exact trait comparison

Figure 1. The exact coefficient bit density and small squared Frobenius- discriminant primes for all 21 seed-level classes. Blue entries were published; orange entries were passed over in the reconstructed streams. The P-384 and P-521 singleton fields appear for completeness but supply no within-field published-versus-omitted comparison. Source: analysis.json from the pinned 21-curve archive.

Exact scope and method

The input is the eleven seed-level classes in assessed.json and ten published controls in traits.json. Their SHA-256 digests are `370e79ce0bff8f26bb437fe37d1756f3f8261a5c00e9062ba41cf70f272e8841` and `a1102bd487379d223581863fab731d6eb6957bf0e264b450f1c7b298279b0b67`. Each record retains the exact prime field, short-Weierstrass coefficients, prime subgroup order, cofactor-one group, seed ordinal, and EC1 UID. The unpublished generators in that archive are study-defined; there is no historically published generator for those curves. The P-224 sign pair is one isomorphism class per seed. Existing curve assessment documents the order, primality, and bounded embedding-degree checks; this round reuses those inputs and recomputes `t`, `t²−4p`, twist order, and the seed-to-coefficient relation before feature extraction.

For every curve, analyze.py trial-divides the absolute Frobenius discriminant by every odd prime $\ell \le 10{,}000$ and records exact valuations. For ordinary curves, $\ell^2 \mid \Delta$ is necessary for odd ℓ to divide the conductor f_π of the Frobenius order; the exact local valuation is $\mathrm{floor}(v_\ell(\Delta)/2)$. This is **not** a certificate for the conductor f_E of the particular representative's endomorphism ring. The study's other fixed features and controls are specified in the protocol. Every value is in analysis.json, with a script and input digest.

Small squared divisor in Δ, $\ell \le 10{,}000$	Omitted	Published
None	10	8
3^2	0	1: P-224
5^2	1: 239-bit block 2682	1: prime239v2

The remaining cofactor after bounded trial division was not factored for each discriminant. Consequently, this table says nothing about squared prime factors above 10,000 or the complete f_π and f_E values. It does show that a small conductor divisor in this range cannot be a property shared by all eleven passed-over classes. Small horizontal isogenies may exist even when this conductor test is negative; the test only addresses conductor change at those primes.

The four fields with both labels contain eight published and eleven omitted curves, yielding 25 within-field published-versus-omitted pairs. The predeclared comparison tests all **4,480** allocations that preserve each field's published count. For coefficient `b` bit weight, the published value is lower in **22/25** pairs. Its two-sided exact allocation tail is **67/4,480**. Across all 21 fixed, correlated features, the exact minimum-tail calibration is **254/4,480**. These are diagnostics of this *retrospective* family: historical labels were not randomly assigned, earlier investigations already exposed the coefficient pattern, and the 21 rows are prefixes ending at published blocks rather than a random population. Neither fraction is a historical probability or a weakness estimate. In P-224, two omissions have lower sign-invariant coefficient bit weight than published P-224; therefore even the leading feature fails as a universal selector.

Completed twist factorizations

FACTOR_PROTOCOL.md fixed the two residuals, 128 MiB GP stack, 300-second cap, output paths, and verification before the new attempts. The exact commands, software versions, stdout, stderr, elapsed wall times, and results are preserved in the two factor receipts. Sage verification, generated by verify_twists.sage, binds both receipts by SHA-256 and checks every new factor with `is_prime(proof=True)` plus the full twist-order products.

Curve / stream block	Small factors	New proven factor sizes	Largest factor
Published prime239v2 / 2320	19×103	82 and 147 bits	147 bits
Omitted 239-bit / 2682	29	95 and 140 bits	140 bits

The exact complete products are:

```
prime239v2 (block 2320):
  19 * 103 * 43467619736112711747293 59
           * 10385138167812959103228894546307 4862302452823

omitted 239-bit curve (block 2682):
  29 * 391556778820688836852884327 61
     * 77799394813092542320477252108 9485491568821
```

The GP factor calls took **15.117** and **29.249** elapsed wall seconds on the recorded host; these are local run observations, not estimates of 1997 cost. The older selection probe had 18/21 complete cases under its 20-second cap; the P-224 follow-up completed one more. The two records above close that specific data gap. Several omitted curves have *larger* largest twist factors than published curves in the same field, including the 192-bit and P-224 examples documented in the selection analysis. A small twist factor has a protocol-specific effect only when unsuitable points can reach a relevant scalar operation; these factorizations do not establish an ordinary-group attack.

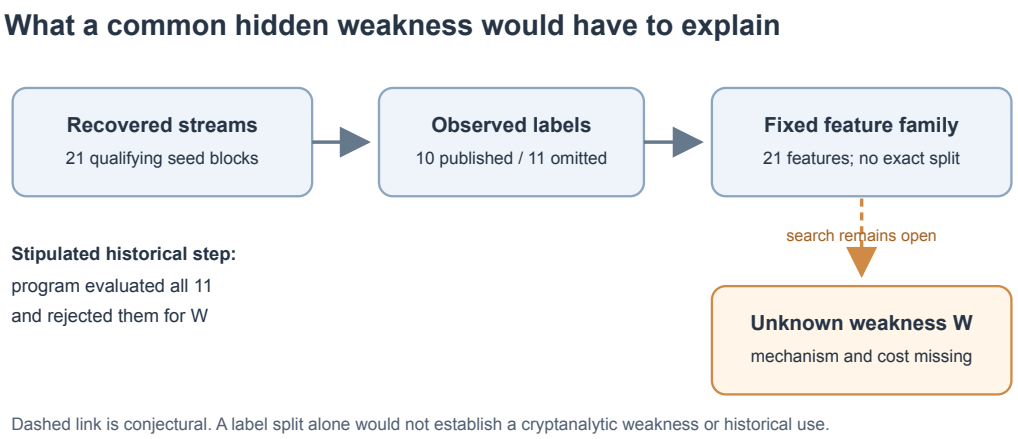


Figure 2: Inference boundary

Figure 2. The user's historical-evaluation and hidden-weakness assumptions are shown as a conjectural branch. The solid arithmetic comparison ends at a bounded negative result; no attack edge is inferred. Editable source: render_visuals.py.

What remains open

Proposed, not established: The eleven curves could fail a multi-part or field-specific criterion, or they could share a mathematical property outside the checked family. A property that reduces unknown-scalar work needs a specific algorithm or transfer, subgroup preservation, matched published controls, full cold and amortized cost, memory, and any withheld witness or public reconstruction cost. A hypothetical costly intermediate advantage would matter even without cheap complete recovery; no such advantage has been identified here. A post hoc predicate fitted to the 21 labels needs a genuinely independent historical decision to become a predictive reconstruction, and provenance of the 1997–98 code or logs to become a claim about what the designers actually used.

The next discriminating work is to recover original generator, point-count, or review artifacts, or freeze one mathematically meaningful candidate weakness `W` and test it on an omitted curve and a same-field published control before expanding to all 21. The named selection problem remains open. The existing selection graphs and canonical ECDLP frontier graphs were checked; this study adds exact trait exclusions and two twist factorizations but no verified map or attack edge, so none is changed.

Reproduction and provenance

```
python3 research/nist_seed_commonality_20261023/analyze.py
python3 research/nist_seed_commonality_20261023/render_visuals.py
sage research/nist_seed_commonality_20261023/verify_twists.sage
pandoc research/nist_seed_commonality_20261023/report.md \
  --metadata-file=research/nist_seed_commonality_20261023/pdf_metadata.yaml \
  --pdf-engine=typst -o research/nist_seed_commonality_20261023/report.pdf
```

The factor attempts are write-once and were already run; see factor_twists.py and the receipts rather than rerunning them over the same paths. Source labels: `internal` for the prior exact curve archive, census, selection analysis, and factor receipts; `derived` for the feature table and allocation calculation; `cross-checked` for the Sage factor verification. Historical context is from the primary FIPS 186-2 generation procedure, Valsorda's candidate account, and Anthropic's reconstruction, all previously retrieved for this study line. This is a search report, not a research-state transition.